

Sheridan

Linear Algebra
MATH15325D

Lecture 1 – Worksheet



Complex numbers

The number a is called the *real part* of z and is denoted by $\operatorname{Re}(z)$, and the number b is called the *imaginary part* of z and is denoted by $\operatorname{Im}(z)$. Thus,

$$\operatorname{Re}(3 + 2i) = 3, \operatorname{Im}(3 + 2i) = 2$$

$$\operatorname{Re}(1 - 5i) = 1, \operatorname{Im}(1 - 5i) = \operatorname{Im}(1 + (-5)i) = -5$$

$$\operatorname{Re}(7i) = \operatorname{Re}(0 + 7i) = 0, \operatorname{Im}(7i) = \operatorname{Im}(0 + 7i) = 7$$

$$\operatorname{Re}(4) = \operatorname{Re}(4 + 0i) = 4, \operatorname{Im}(4) = \operatorname{Im}(4 + 0i) = 0$$

Complex number conjugate

Find the conjugate and modulus for each of these numbers

$$z = i, \quad z = 1, \quad z = -4 - 5i, \quad z = 3 + 4i$$

Graphical representation of complex numbers

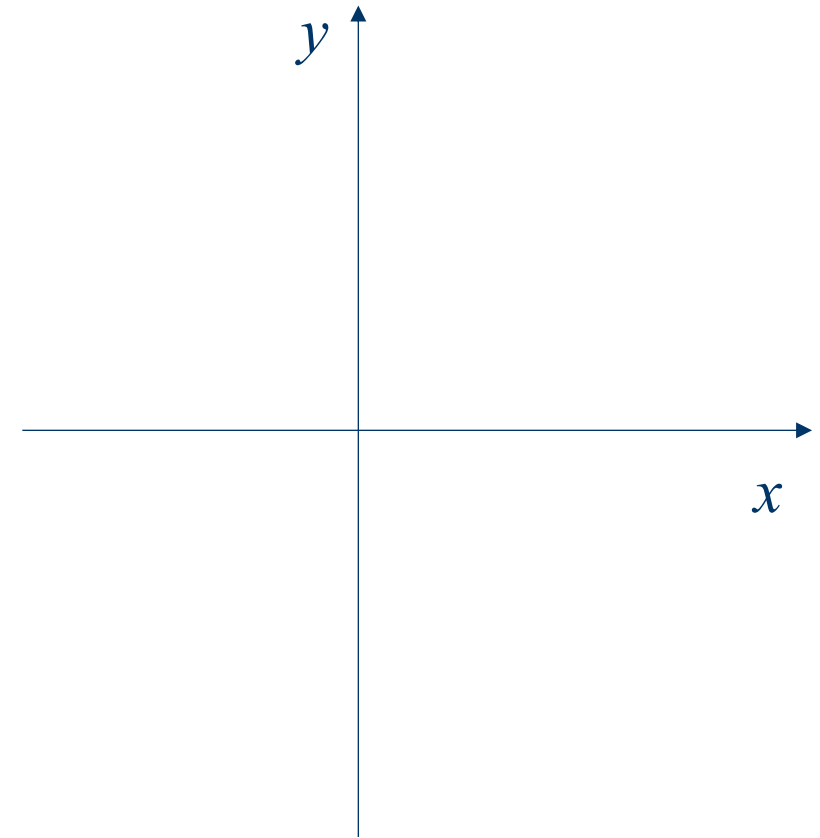
Show the numbers below on the complex plane

$$z_1 = 3 + 4i, \quad z_2 = 3 - 4i, \quad z_3 = -3 + 4i, \quad z_4 = -3 - 4i$$

$$z_5 = 1, \quad z_6 = -1, \quad z_7 = i, \quad z_8 = -i,$$

$$z_9 = 1 + i, \quad z_{10} = 1 - i, \quad z_{11} = -1 + i, \quad z_{12} = -1 - i$$

$$z_{13} = 3 + 4i \quad \text{and} \quad z_{14} = 4 + 3i$$



Multiplication and division

Find the multiplication of the complex numbers below $z_1 z_2$ and
division $\frac{z_1}{z_2}$

$$z_1 = 2 - 5i \quad , \quad z_2 = 4 + 3i$$

$$z_1 = \sqrt{5} - 5i \quad , \quad z_2 = 2 + i$$

Theorem 1 and 2

Using $z_1 = 2 - 5i$, and $z_2 = 4 + 3i$, explore the application of the properties below

THEOREM 1 The following results hold for any complex numbers z , z_1 , and z_2 .

- 1) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
- 2) $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$
- 3) $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$
- 4) $\overline{z_1 / z_2} = \bar{z}_1 / \bar{z}_2$
- 5) $\bar{\bar{z}} = z$

THEOREM 2 The following results hold for any complex numbers z , z_1 , and z_2 .

- 1) $|\bar{z}| = |z|$
- 2) $|z_1 z_2| = |z_1| |z_2|$
- 3) $|z_1 / z_2| = |z_1| / |z_2|$
- 4) $|z_1 + z_2| \leq |z_1| + |z_2|$

Polar form

Express $z = 1 - \sqrt{3}i$ in polar form using the principal argument.

Solution The modulus of z is

$$|z| = \sqrt{1^2 + (-\sqrt{3})^2} = \sqrt{4} = 2$$

Thus, it follows from (10) with $a = 1$ and $b = -\sqrt{3}$ that

$$1 = 2 \cos \phi \quad \text{and} \quad -\sqrt{3} = 2 \sin \phi$$

and this implies that

$$\cos \phi = \frac{1}{2} \quad \text{and} \quad \sin \phi = -\frac{\sqrt{3}}{2}$$

The unique angle ϕ that satisfies these equations and whose radian measure satisfies (12) is $\phi = -\pi/3$ (Figure B.7). Thus, a polar form of z is

$$z = 2\left(\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right)\right) = 2\left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}\right)$$

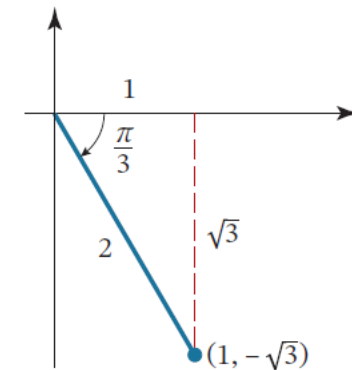


FIGURE B.7

Do the same conversion to $z_1 = 1 + \sqrt{3}i$ and $z_2 = \sqrt{3} + i$

Addition, subtraction, and multiplication

Using the numbers: $z_1 = 2 + 2i$, $z_2 = 1 - \sqrt{3}i$

Find the following amounts in polar forms

$$z_1 + z_2$$

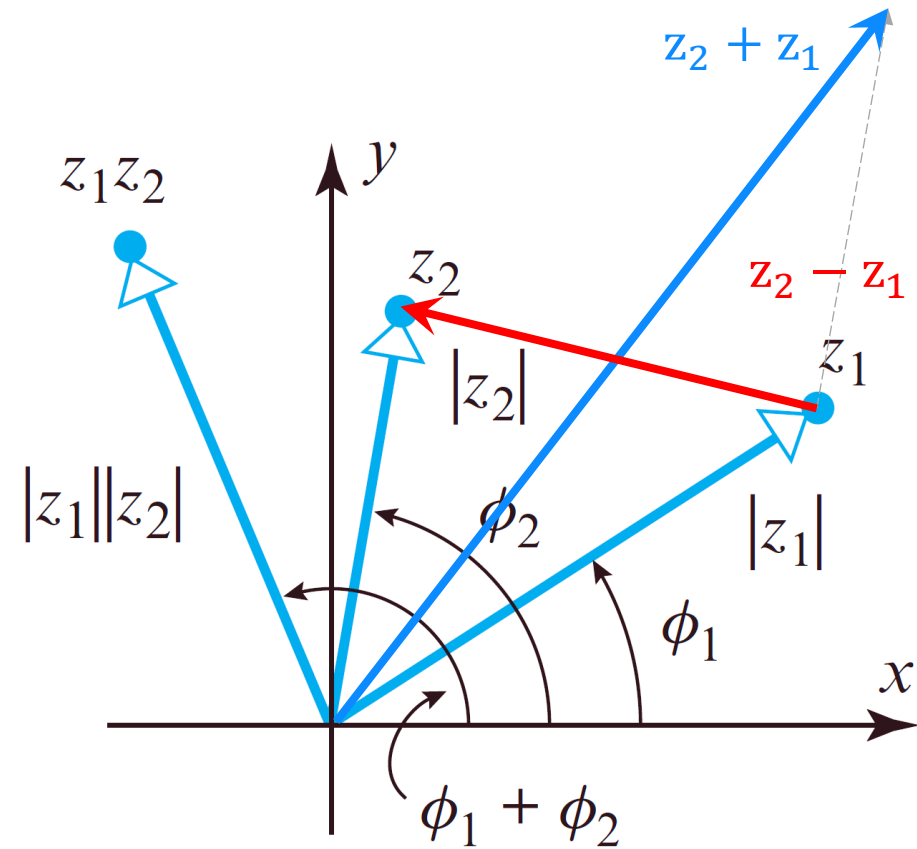
$$z_2 - z_1$$

$$z_1 z_2$$

$$\frac{z_1}{z_2}$$

$$z_2$$

$$iz_1$$



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De Moivre's Formula

Using De Moivre's Formula, find the following:

$$z_1 = (1 + i)^3, \quad z_2 = (1 - \sqrt{3}i)^2$$

$$z_3 = (1 + i)^{1/3}, \quad z_4 = \sqrt{1 - \sqrt{3}i}$$

$$z_5 = \sqrt{(1 - \sqrt{3}i)^5}, \quad z_6 = \sqrt[3]{-1}$$

Euler Formula

Find the following amounts using Euler formula:

1) $e^{i\pi/2} + e^{i\pi}$

2) $e^{2+i\pi}$

3) $e^{2+i\pi} e^{3+\frac{i\pi}{3}}$

4) $e^{2+i\pi} / e^{3+\frac{i\pi}{3}}$